

Revised Generation Interconnection System Impact Study Report

For Queue Project Q19-1350

West Mount Vernon 138 kV
November 21, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1350. The Interconnection Customer proposes a 47.7 MW solar photovoltaic facility interconnecting at the West Mount Vernon 138 kV substation in the Kentucky area. The request was received on December 08, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 509 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 4 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$394,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Tristate Substation 345/1 kV Auto	Hubbardstown / Tristate 345 kV Line	105.8	512	6.7
Tristate 138 kV/1 kV Auto	Tristate / Hubbardstown 345 kV line	103.9	522	6.7
Northwest Beloit 345/1kV TX	Tristate / Valemount 345 kV	100.3	513	6.3
Northwest Beloit 230/1kV Auto	Valemount / Tristate 345 kV	100.3	513	6.3

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Tristate 345 kV/1 kV Transformer Bank	Replace transformer with higher-rated unit	\$9,867,000	1% (\$121,000)
Tristate 138 kV/1 kV TX	Replace transformer with higher-rated unit	\$3,317,000	1% (\$41,000)
Northwest Beloit 345/1kV TX	Replace transformer with higher-rated unit	\$14,924,000	1% (\$171,000)
Northwest Beloit 230/1kV Auto	Replace transformer with higher-rated unit	\$5,350,000	1% (\$61,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
	Total allocated to project		\$394,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.